Band-1 Hadamard Order 5444 — Miyamoto (Kronecker H_4×H_340 Paley-II GF(13^2) + C_1362); 5444 IN SPK via Paley-I; Gap 2052 Still OPEN

H_340/H_1360/C_1362/H_5444 Miyamoto PASS; H_5444 Paley-I PASS (SPK); NOT Miyamoto-only; gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-09-15

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_5284_Miyamoto`; `2026-09-15_Band1_Hadamard_5188_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_5444_miyamoto/`.
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Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 340, 1360 and 5444 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Kronecker seed $H_{1360} = \text{Kronecker}(H_4, H_{340})$ where $H_{340} = \text{Paley-II}(q = 169 = 13^2)$ over GF(13^2), plus independent SPK path $H_{5444} = \text{Paley-I}(q = 5443)$. Seed triage for H_{1360}: Paley-II / Double-Paley-I / Paley-I-direct all **FAIL**. It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. Order **5444 is IN SPK via Paley-I** ($q = 5443$ prime $\equiv 3 \pmod 4$); Miyamoto is an **alternate construction**, **NOT** a Miyamoto-only bonus. Paley-II for 5444 FAIL ($q = 2721 = 3 \cdot 907$); Kronecker FAIL (4×1361). `clay_closed = false`.

Abstract

Finite constructions of orders **340**, **1360**, and **5444** via Miyamoto + Paley-I SPK path.

- H_{340} = Paley-II ($q = 169 = 13^2$, GF(13^2))
- H_4 = Sylvester
- H_{1360} = Kronecker(H_4, H_{340})
- C_{1362} = Paley conference ($q = 1361$)
- H_{5444} Miyamoto = Miyamoto(H_{1360}, C_{1362})
- H_{5444} Paley-I = Paley-I ($q = 5443$) — **SPK path**

SPK: 5444 in SPK via Paley-I (NOT Miyamoto-only). Seeds 340/1360 enter SPK via Paley-II prime power / Kronecker.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_5444.py`

Results. overall_pass=true; units 81/81; wall 182.8953s; all max|HH −nI|=0.

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Arithmetic

- Target $n = 5444 = 4 \cdot 1361$; 1361 prime $\equiv 1 \pmod 4$ → Miyamoto-eligible.
- Need H_{1360} and C_{1362} (Paley conference over prime 1361).
- Seed: Kronecker(H_4, H_{340}) with H_{340} = Paley-II $q = 169 = 13^2$.
- Triage FAIL for 1360: Paley-II $q = 679 = 7 \cdot 97$; Double half=680; Paley-I direct $q = 1359 = 3^2 \cdot 151$.
- SPK: Paley-I $q = 5443$ prime $\equiv 3 \pmod 4$ **ELIGIBLE** → verified → **in SPK**.

Constructions (verified)

Object	Method	max|HH −nI|
H_4	Sylvester	0
H_340	Paley-II GF(13^2) q=169	0
H_1360	Kronecker(H_4, H_340)	0
C_1362	Paley conference q=1361	0
H_5444 Miyamoto	Miyamoto(H_1360, C_1362)	0
H_5444 Paley-I	Paley-I q=5443 (SPK)	0

Honesty

- clay_closed = false

- GATE-04 = LOCKED
- Hadamard conjecture = OPEN
- Gap 2052 = OPEN (method triage / non-finding ONLY)
- 5444_in_spk = true (via Paley-I)
- 5444_miyamoto_only = false